

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

PROJECT ROLES & RESOURCE ALLOCATION

1. Project Developer Profile

The OptiCrop project was developed entirely as an individual project. The developer executed all roles across the software development lifecycle:

Development Role	Responsibilities Executed	Key Deliverables Completed
Product Lead	Requirements elicitation, planning, and documentation.	SRS, empathy mapping, and user journey maps.
ML Engineer	Data preprocessing, algorithm selection, and training.	Pickled model pipelines and evaluation plots.
Backend Developer	Flask server controller logic, route validation, database logger.	app.py, user_predictions.csv, health check routes.
UI/UX Designer	Glassmorphic dashboard style guidelines, responsive layouts.	index.html, result.html, dashboard.html , templates.